Tukey's EDA Philosophy

Workshop 4 — Module 1

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Data Visualization for Data Scientists
UNYT Tirana

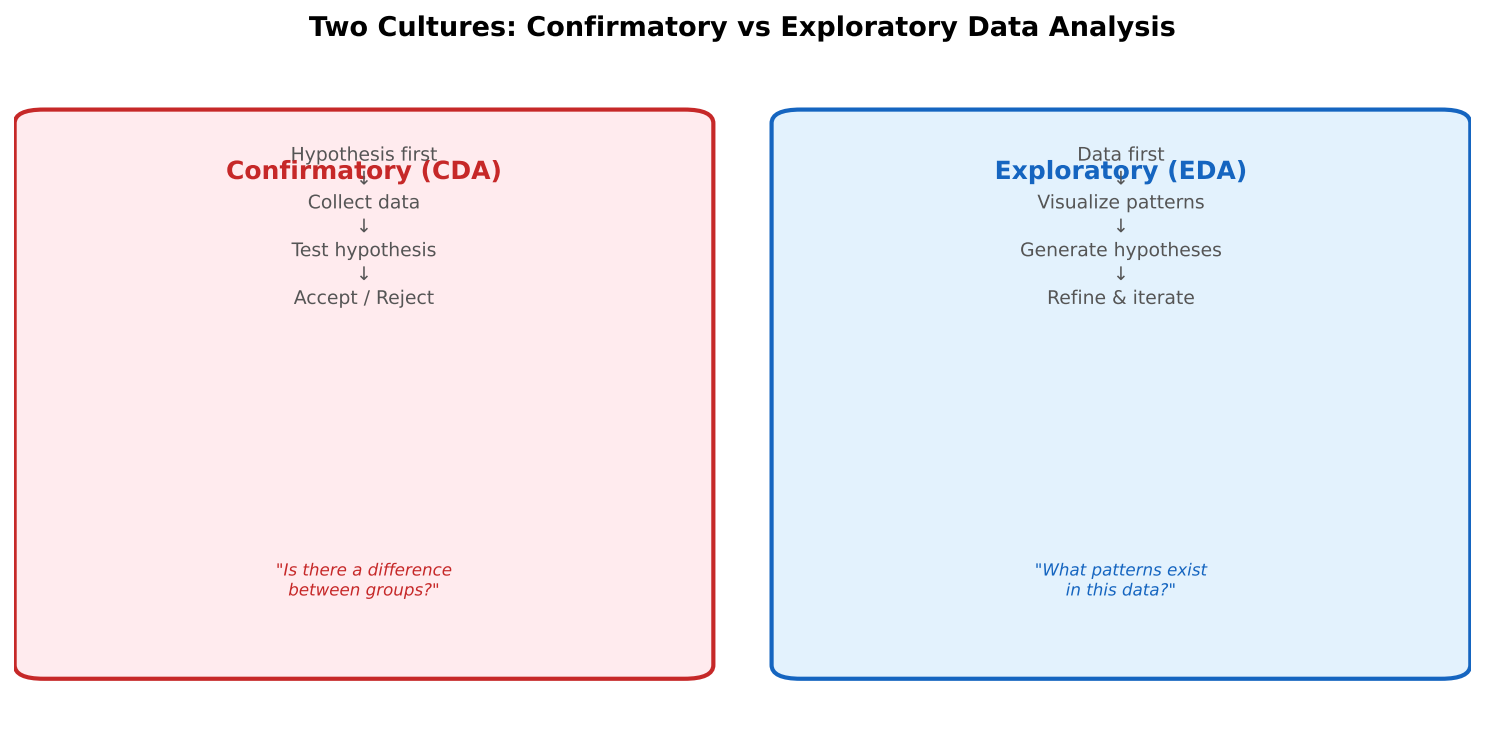
Module Roadmap

1. Confirmatory vs exploratory: two cultures of data analysis
2. Tukey's detective metaphor (1977)
3. The original EDA toolkit: stem-and-leaf, boxplot, residuals
4. Letter-value plots: extending the box for large n
5. The 4-plot diagnostic panel
6. The modern EDA workflow

Learning Outcome
You will understand EDA as a philosophy (not a technique), adopt Tukey's iterative "look at the data first" mindset, and apply the 4-phase EDA workflow to any new dataset.

Two Cultures

Confirmatory vs Exploratory



Pro-Tip

Most statistics courses teach CDA: hypothesis $\rightarrow$ test $\rightarrow$ p-value. Tukey argued that this skips the most important step: **looking at the data**. EDA generates the hypotheses that CDA then tests. Both are necessary; EDA comes first.

Tukey's Own Words

"Exploratory data analysis is detective work — numerical detective work — or counting detective work — or graphical detective work."

— John W. Tukey, *Exploratory Data Analysis* (1977)

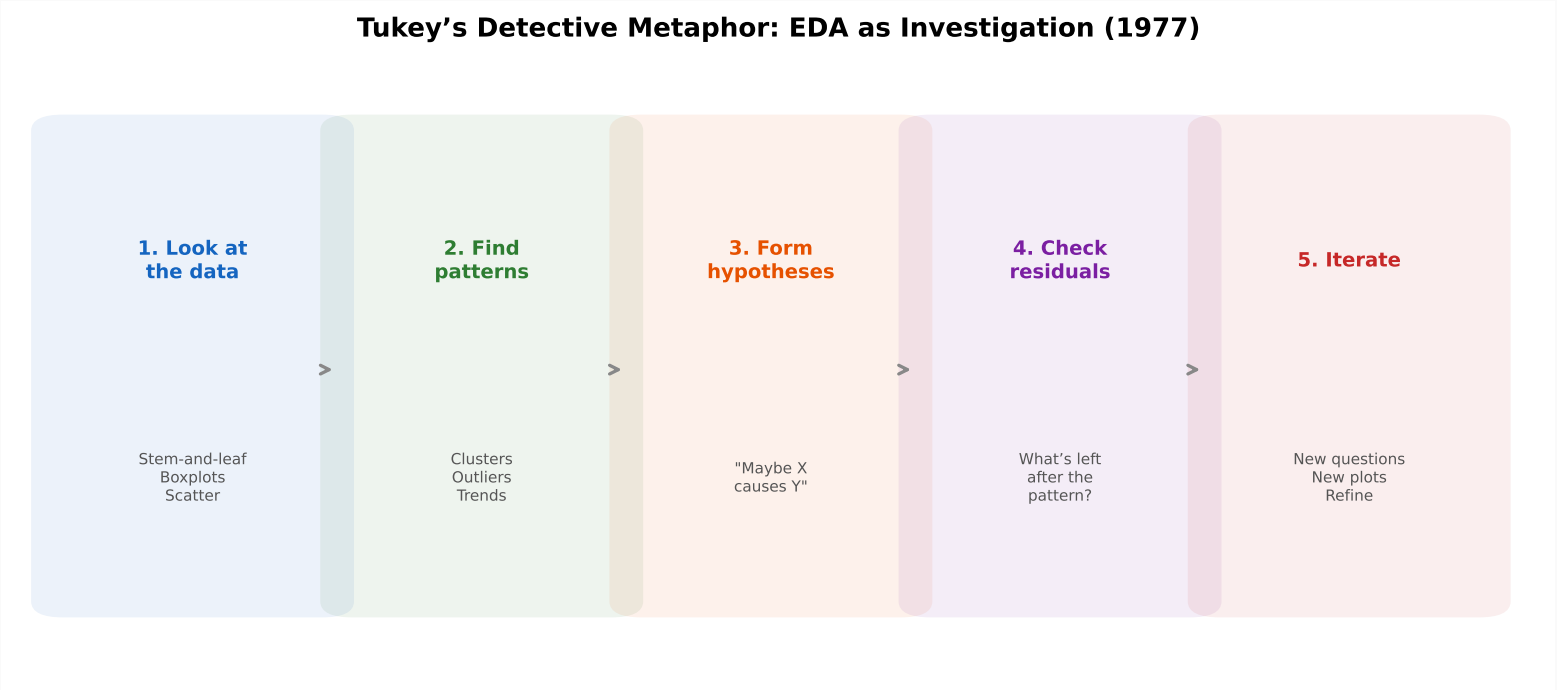
"The greatest value of a picture is when it forces us to notice what we never expected to see."

— John W. Tukey

Data Insight
Tukey invented the boxplot, the stem-and-leaf display, the term "bit" (with Shannon), and the FFT algorithm (with Cooley). His 1977 book *Exploratory Data Analysis* founded the field.

The Detective Metaphor

EDA as Investigation: Five Steps



Key Principles of EDA

Principle	Implication
Scepticism	Assume nothing about the data; let it speak
Iteration	One plot leads to the next question
Robustness	Use medians over means; resist outlier influence
Residuals	After fitting a pattern, examine what remains
Graphical primacy	Tables summarise; plots reveal
Re-expression	Transform to simplify (log, sqrt, reciprocal)

Tukey's Original Toolkit

Stem-and-Leaf: Every Value + Shape

Stem-and-Leaf: Every Value Visible + Shape

Stem	Leaves
3	0 2 2 6 6 7 7
4	2 3 4 7 7 8
5	0 0 4 4 6 7 7 7 8 9
6	1 2 2 2 2 8
7	2

n = 30, stem unit = 10

Data Insight

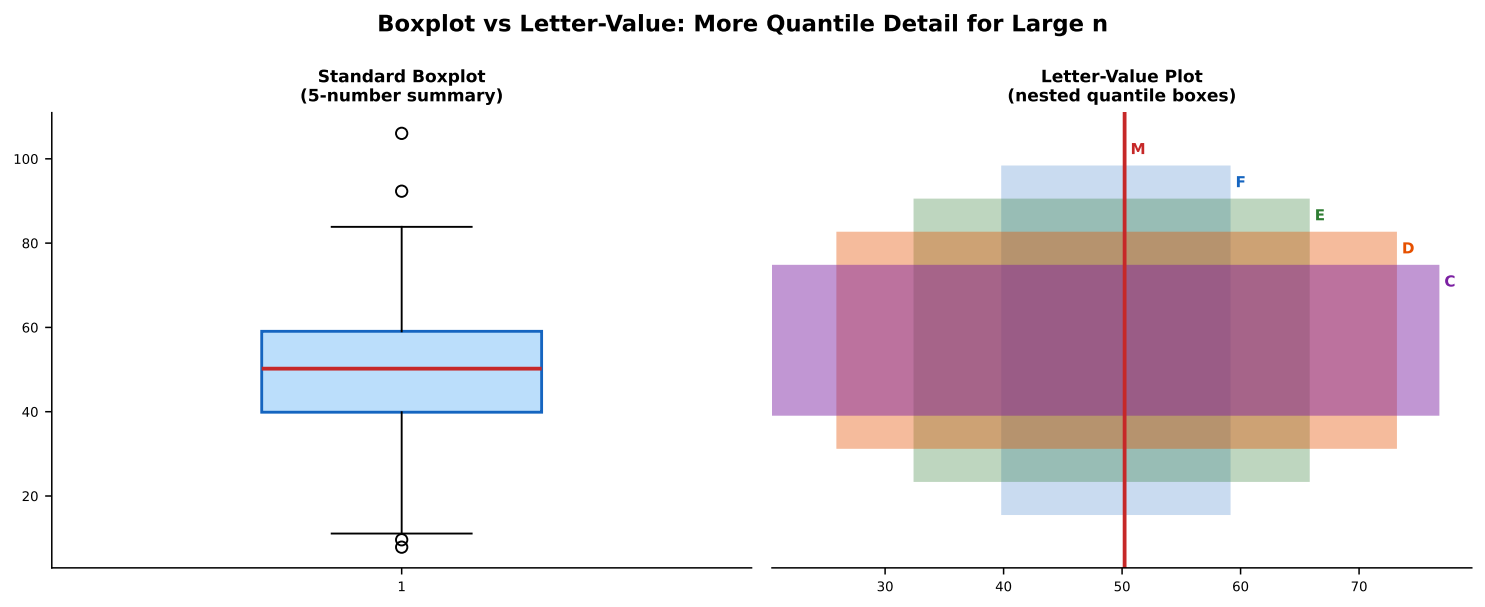
The stem-and-leaf preserves every data value (unlike a histogram which bins) while showing the distribution shape. Modern equivalent: `stem()` in R, or simply a high-resolution histogram. Useful for $n < 100$ in teaching contexts.

From Tukey (1977) to Modern R/Python

Tukey's Original EDA Toolkit (1977) → Modern Equivalents

Method	Purpose	Modern R Code	Status
Stem-and-leaf	Every value + shape	`print() / stem()`	Replaced by histogram
Box-and-whisker	5-number summary	`geom_boxplot()`	Still core EDA tool
Residual plots	What's left after fit	`plot(lm_model)`	Foundation of diagnostics
Median polish	Robust 2-way table	`medpolish()`	Additive decomposition
Re-expression	Transform (log, sqrt)	`scale_x_log10()`	Stabilise variance
Smoothing	Running medians	`geom_smooth()`	Modern: LOESS, GAM

Letter-Value Plot: Beyond the 5-Number Summary

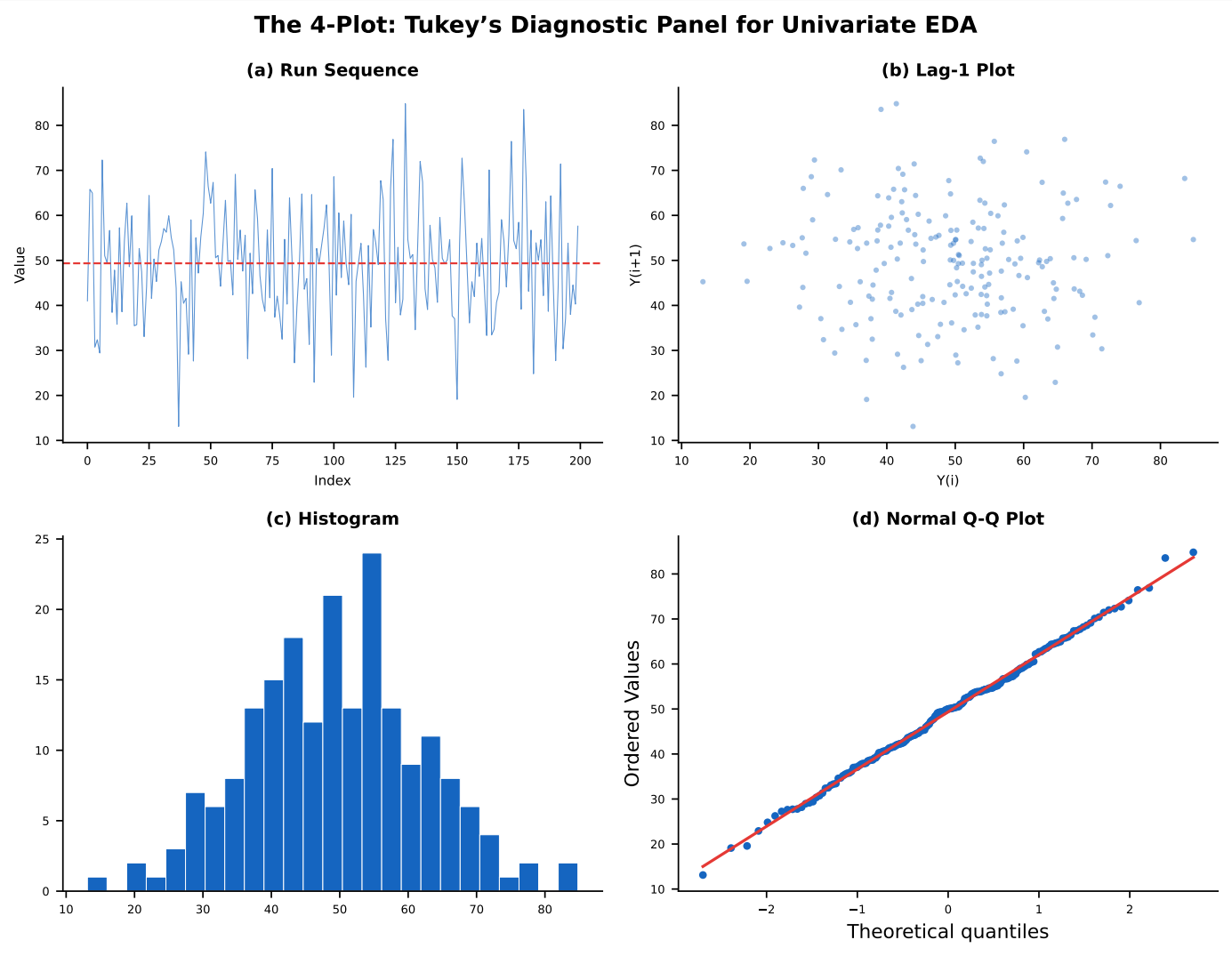


Pro-Tip

For large n (> 200), the standard boxplot wastes information: it shows only 5 quantiles. The **letter-value plot** (Hofmann et al., 2017) adds nested boxes at the eighths (E), sixteenths (D), etc., revealing tail behaviour. In R: `ggplot2::geom_lv()`. In Python: `sns.boxenplot()`.

The 4-Plot

Tukey's 4-Plot Diagnostic Panel



What Each Panel Reveals

Panel	What It Checks
(a) Run sequence	Are there shifts, trends, or cycles over time/index?
(b) Lag-1 plot	Is there autocorrelation? (If cloud $\approx$ circle: no)
(c) Histogram	What is the shape? Unimodal, skewed, bimodal?
(d) Normal Q-Q	Do the data follow a normal distribution? Departures at tails?

If (a) shows no trend, (b) shows no autocorrelation, (c) looks unimodal, and (d) follows the diagonal — the data are well-behaved for parametric analysis. Deviations guide further EDA.

The 4-Plot in R vs Python

R

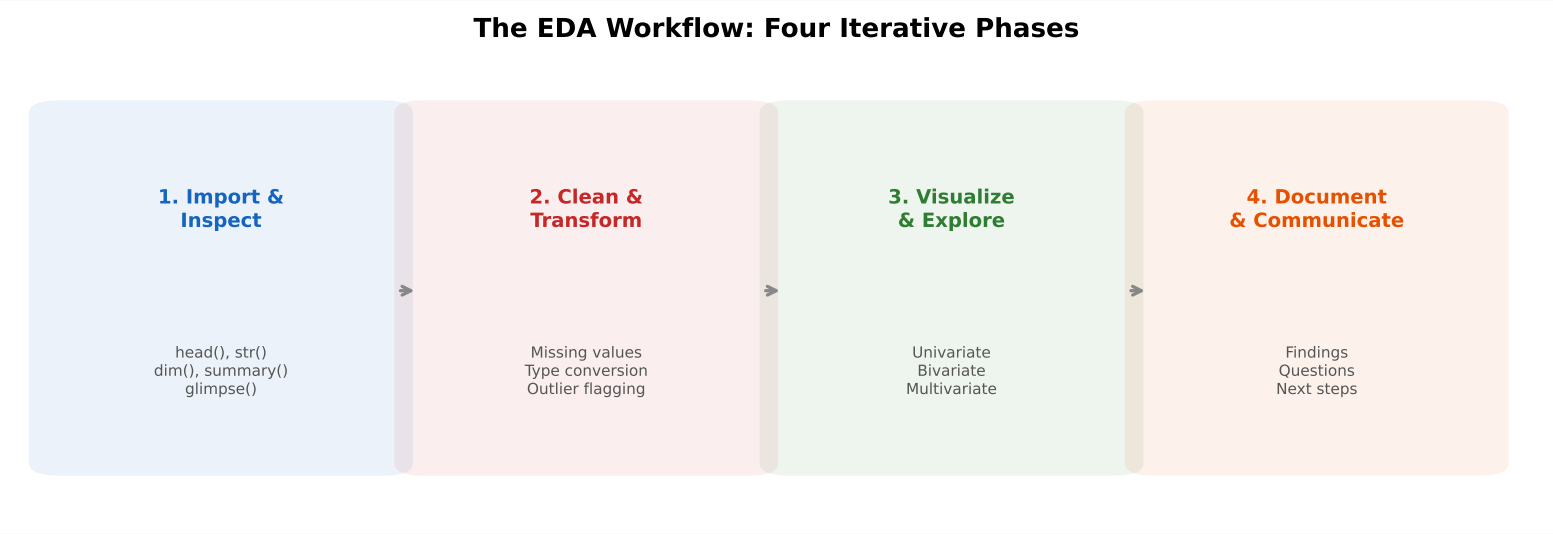
```r
library(patchwork)

p_run <- ggplot(df, aes(x = 1:n()
    ,
    y = value)) + geom_line() +
  geom_hline(yintercept = mean(
    df$value), color = "#E53935",
    linetype = "dashed") +
  labs(title = "(a) Run Sequence"
      )

p_lag <- ggplot(df, aes(
    x = lag(value), y = value)) +
  geom_point(alpha = 0.3) +
  labs(title = "(b) Lag-1 Plot")

p_hist <- ggplot(df, aes(
    x = value)) +
  geom_histogram(bins = 25) +
  labs(title = "(c) Histogram")

p_qq <- ggplot(df, aes(
    sample = value)) +
  geom_qq() + geom_qq_line(
```

Python

```python
from scipy.stats import probplot

fig, axes = plt.subplots(2, 2)

# (a) Run sequence
axes[0,0].plot(range(n), values)
axes[0,0].axhline(
    y=np.mean(values),
    color="#E53935", ls="--")
axes[0,0].set_title("(a) Run")

# (b) Lag-1
axes[0,1].scatter(
    values[:-1], values[1:],
    s=8, alpha=0.3)
axes[0,1].set_title("(b) Lag-1")

# (c) Histogram
axes[1,0].hist(values, bins=25)
axes[1,0].set_title("(c) Hist")

# (d) Q-Q
probplot(values, dist="norm",
    plot=axes[1,1])
```

Modern EDA Workflow

The Four-Phase EDA Workflow



Phase 1: Import & Inspect

R

```r
library(tidyverse)

df <- read_csv("data.csv")
dim(df)              # rows x cols
glimpse(df)          # types +
    preview
summary(df)          # 5-num + NA
    count

# Comprehensive EDA report
library(skimr)
skim(df)              # grouped
    stats

# Or auto-generated report
library(DataExplorer)
create_report(df)   # HTML report
```

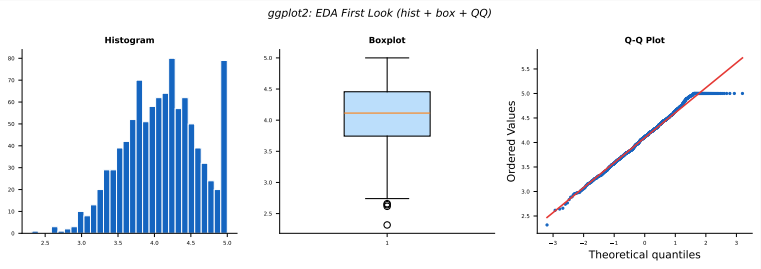
Python

```python
import pandas as pd

df = pd.read_csv("data.csv")
df.shape              # (rows, cols)
df.info()             # types + non-
    null
df.describe()         # stats for
    numeric
df.head()             # first 5 rows

# Comprehensive EDA
# pip install ydata-profiling
from ydata_profiling import (
    ProfileReport)
report = ProfileReport(df)
report.to_file("eda_report.html")
```

Complete Examples

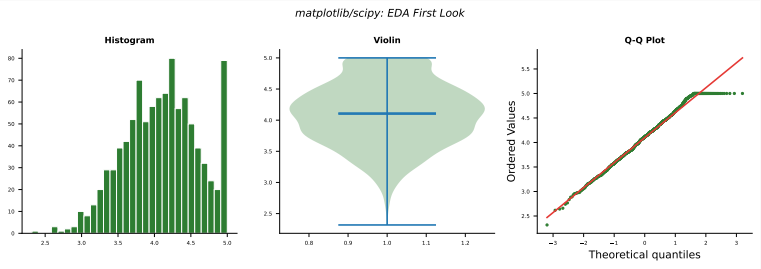
R: EDA First Look

```r
library(tidyverse); library(
    patchwork)
apps <- read_csv(
  "googleplaystore.csv") |>
  filter(!is.na(Rating))

p1 <- ggplot(apps, aes(x=Rating))+
  geom_histogram(bins=30,
    fill="#1565C0", color="white")
        +
  theme_minimal() +
  labs(title="(a) Histogram")

p2 <- ggplot(apps, aes(x="",
    y=Rating)) +
  geom_boxplot(fill="#BBDEFB",
    width=0.3) +
  theme_minimal() +
  labs(title="(b) Boxplot", x=NULL)

p3 <- ggplot(apps,
    aes(sample=Rating)) +
  geom_qq(alpha=0.1, size=0.5) +
  geom_qq_line(color="#E53935") +
  theme_minimal() +
  labs(title="(c) Q-Q Plot")
```



Python: EDA First Look

```python
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import probplot

apps = pd.read_csv(
    "googleplaystore.csv")
ratings = apps["Rating"].dropna()

fig, axes = plt.subplots(1, 3,
    figsize=(10, 3.5))

axes[0].hist(ratings, bins=30,
    color="#2E7D32",
    edgecolor="white", lw=0.3)
axes[0].set_title("Histogram")

parts = axes[1].violinplot(
    [ratings],
    showmeans=True,
    showmedians=True)
axes[1].set_title("Violin")

probplot(ratings, dist="norm",
    plot=axes[2])
axes[2].set_title("Q-Q Plot")
```



Synthesis

Key Takeaways

1. **EDA is a philosophy**, not a technique: look at the data before modelling. CDA tests hypotheses; EDA generates them.
2. Tukey's **detective metaphor**: investigate, form hypotheses, check residuals, iterate. Never trust a single plot.
3. The **4-plot** (run sequence + lag-1 + histogram + Q-Q) is the first diagnostic panel for any univariate analysis.
4. **Letter-value plots** extend boxplots for large n by showing nested quantile boxes: `geom_lv()` or `sns.boxenplot()`.
5. The **modern EDA workflow**: Import/Inspect $\rightarrow$ Clean/Transform $\rightarrow$ Visualize/Explore $\rightarrow$ Document/Communicate.
6. Automate Phase 1 with `skimr::skim()` (R) or `ydata-profiling` (Python) for instant EDA reports.

Essential Readings

- Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley. (The founding text.)
- Hofmann, H., Wickham, H. & Kafadar, K. (2017). "Letter-value plots." *Journal of Computational and Graphical Statistics*.
- Wickham, H. & Grolemund, G. (2023). *R for Data Science*, 2nd ed., Chapters 2, 10 (Workflow, EDA).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 7 (Histograms and Density).

Next Module: Data Wrangling for Visualization in R (W04-M02)